

Two numbers for a whole set of beliefs: what PLN’s strength and confidence can and cannot see

Mettapedia Working Notes

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Abstract

A belief in PLN (Probabilistic Logic Networks) carries two numbers: a *strength* s and a *confidence* c . This paper explains, from scratch, what those two numbers really stand for and exactly when they tell the whole story. The short version: behind the pair $\langle s, c \rangle$ sits a whole *set* of probability assignments you have not yet ruled out; strength is the centre of that set on the question you asked, and confidence measures how narrow the set is. Two numbers are a faithful summary of the set when the set pins your question down to a single value, and a lossy summary when the set leaves a genuine *interval* of admissible answers.

We make this precise with a single recurring picture — *pinned* versus a *range* — and follow it up three rising levels: one belief; a belief built from many local pieces; and three infinite limits (order-free data streams; weakly coupled systems, which collapse to one precise answer; and strongly coupled systems, where distant conditions keep a real interval open). Along the way we catalogue nineteen design choices — “degrees of freedom” — that two numbers either expose or hide, each with a numeric example you can run. Every claim has a machine-checked counterpart; the formal statements are collected, de-jargoned, in an appendix, and a runnable example pack reproduces every number in the paper.

Contents

1 One flip versus a hundred

You flip a bent coin once and it lands heads. *One out of one — so far it always comes up heads.* Now I flip a different bent coin a hundred times and it lands heads every single time. *A hundred out of a hundred.* In both cases the coin “always comes up heads” on the evidence so far, so the best guess for the next flip is the same: lean all the way toward heads. But you are not equally sure. After one flip you would shrug; after a hundred you would bet real money.

That gap is the whole subject of this paper. The “best guess” — which way the evidence leans — is the **strength**. “How much you would bet on your own guess” — how much evidence stands behind it — is the **confidence**. They are genuinely two different things, and *no single number can be both*: the two coins above share a strength but must not share a confidence.

Here is the idea that organizes everything below. Behind those two numbers sits a richer object: **the whole range of probabilities still consistent with what you have seen**. After 1/1 that range is wide — the coin’s true bias could be anywhere from “barely favours heads” to “always heads.” After 100/100 the range has shrunk to a sliver near 1. PLN keeps only two facts about that range: where its centre is (strength) and how narrow it is (confidence). So the question of this paper is:

When do those two numbers tell the whole story, and when is the range behind them doing something the two numbers cannot show?

One more twist before we start, to set expectations. Even the *rate* at which confidence grows is a choice. The same five pieces of evidence can be “a lot” or “barely anything” depending on the domain. Using PLN’s standard rule $c = N/(N + \kappa)$ with $N = 5$ pieces of evidence: at $\kappa = 1$ you get confidence 0.833; at $\kappa = 10$, only 0.333; at $\kappa = 50$, just 0.091. Same evidence, three confidences. That dial κ — and roughly eighteen of its cousins — is what Section ?? catalogues. Each is a place where two beliefs can look identical as $\langle s, c \rangle$ yet mean different things; that is what we mean by a *degree of freedom*.

What this paper is, and is not. We characterize a single truth value: what $\langle s, c \rangle$ records, when it is forced, and what it hides. We do *not* re-derive how truth values propagate through PLN’s inference rules (deduction, induction, abduction, revision); that is the existing PLN corpus (ξ PLN, ν PLN, and the WM-PLN book [?, ?, ?]). The body stays at sophomore level: basic probability (a coin’s bias, expectation, conditional probability) is all you need. The rigorous statements live in Appendix ??, and a runnable companion that reproduces every number lives at [papers/pln-confidence-examples/](https://github.com/zariuq/pln-confidence-examples/) (Appendix ??). All sources are in the public MeTTapedia repository, <https://github.com/zariuq/MeTTapedia>, and the file references throughout this paper link to it.

2 Two dials: which way, and how much

Picture a belief as a little instrument panel with two dials (Figure ??).

The strength dial reads the ratio. Count your evidence as n^+ votes “for” and n^- votes “against.” Strength reads which way the votes lean:

$$s = \frac{n^+}{n^+ + n^-}.$$

For a coin, this is just the fraction of heads. 7 heads in 10 flips gives $s = 0.7$.

The confidence dial reads the total. Confidence ignores which way the votes went and reads *how many* there are. Writing $N = n^+ + n^-$ for the total evidence,

$$c = \frac{N}{N + \kappa},$$

where $\kappa > 0$ is a calibration constant for “how much evidence counts as a lot.” More evidence pushes c toward 1; no evidence gives $c = 0$.

Why two dials and not one? Because the two facts vary independently. Consider:

evidence	strength s	total N	confidence c (at $\kappa = 1$)
1 for, 1 against	0.5	2	0.667
10 for, 10 against	0.5	20	0.952
100 for, 100 against	0.5	200	0.995
1 for, 1 against	0.5	2	0.667
2 for, 0 against	1.0	2	0.667

The top three rows have the *same* strength 0.5 but climbing confidence: balanced evidence still accumulates certainty about “it’s a fair coin.” The bottom two have the *same* confidence (same total $N = 2$) but different strength. Any attempt to fold belief into a single number would have to equate 1/1 with 100/100 — exactly the mistake the opening coins warn against.

The calibration dial κ . The constant κ re-gears the confidence dial without touching strength. With the same $N = 5$:

$$\kappa = 1 : c = 0.833 \quad \kappa = 10 : c = 0.333 \quad \kappa = 50 : c = 0.091.$$

A hand-curated clinical fact (trust it fast: small κ) and a single noisy high-throughput hit (trust it slowly: large κ) can have identical $+/-$ ratios and still deserve very different confidence. κ is a knob you set per domain; nothing in the mathematics picks it for you. (We will see in Section ?? that, once you fix a couple of natural conventions, the *shape* $c = N/(N + \kappa)$ is forced¹ — but the value of κ never is.)

3 Behind the dials: a set of admissible probabilities

We can say what the dials are really reading. Suppose your belief is about some quantity — “the coin lands heads,” or more generally the expected value

¹Lean: `walley_width_complement_forces_plnOdds` in [Mettapedia/Logic/PLNConfidenceWeight.lean](#); see Appendix ??.

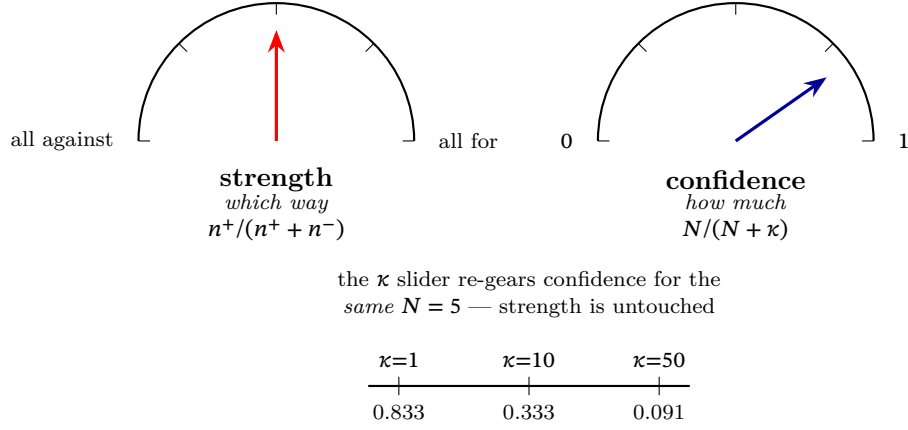


Figure 1: Two dials. Strength reads which way the evidence leans (the ratio); confidence reads how much evidence there is (the total), geared by the calibration constant κ . The two move independently — which is why one number cannot do the job of both.

of some measurement X . If you knew the exact probability distribution of the world, X would have one exact expected value. But usually you do *not* know the exact distribution; you know a *set* of distributions that are all consistent with what you have seen. That set is the object behind the two numbers. In the literature it is called a **credal set**; we will also just call it the *spread* of admissible probabilities.

The whole content of $\langle s, c \rangle$.² Take the spread. Ask your question X . Each admissible distribution in the spread assigns X some expected value; sweeping over the spread, those values range between two endpoints, L the smallest expected value anyone in the spread allows and U the largest. (For the convex spreads we care about they fill the whole interval $[L, U]$; but only the two endpoints are ever read off.) Then

$$s = \frac{1}{2}(L + U) \quad (\text{the centre of the interval}),$$

$$c = g(U - L) \quad (\text{a decreasing function of its width}).$$

with $g(0) = 1$ (zero width: full confidence) and g falling toward 0 as the interval widens (maximum width: no confidence).

That is the entire “confidence characterization” in one box, and the rest of the paper is unpacking it. The endpoints L and U are the smallest and largest

²The read-off: `lowerEnvelope` and `upperEnvelope` in [Mettapedia/ProbabilityTheory/ImpreciseProbability/ProjectiveCredal.lean](#); see Appendix ??.

honest answers; strength sits in the middle, and confidence asks how far apart they are (Figure ??).

This is the same confidence as Section ??. It is not a second, competing definition. Take g to be one-minus-width (the natural “width-complement” convention; Section ??, entry 5). After N observations, the range of biases still consistent with the data has width $\kappa/(N + \kappa)$, so $1 - \text{width} = N/(N + \kappa)$ — exactly the confidence dial of Section ?? . The count rule $c = N/(N + \kappa)$ and the envelope-width rule $c = g(U - L)$ are one belief seen two ways: counting evidence *is* narrowing the spread.

The coin, concretely. Suppose all you know is that the coin’s bias p lies somewhere in $[0.4, 0.6]$. The spread is the family of coins with those biases. The expected value of “heads” is just p , so it ranges over $[0.4, 0.6]$: here $L = 0.4$, $U = 0.6$. Strength $= \frac{1}{2}(0.4 + 0.6) = 0.5$; confidence $= 1 - 0.2 = 0.8$ (reading g as one-minus-width). Now suppose you learn nothing and your spread is all of $[0, 1]$: then $[L, U] = [0, 1]$, width 1, and confidence drops to 0 — you have a centre (0.5) but no reason to trust it. Shrink the spread to the single bias $\{0.5\}$ and the interval collapses to the point 0.5, width 0, confidence 1.

Why a set, and why two numbers lose information. The spread is richer than any two numbers: it can have a complicated shape, and it can pin some questions while leaving others wide open. Recording only the centre and width of one question’s interval throws the rest away. That lost information is not a bug — it is the price of a compact, composable truth value — but it is exactly the source of the “degrees of freedom” this paper is about (Figure ??, grey box).

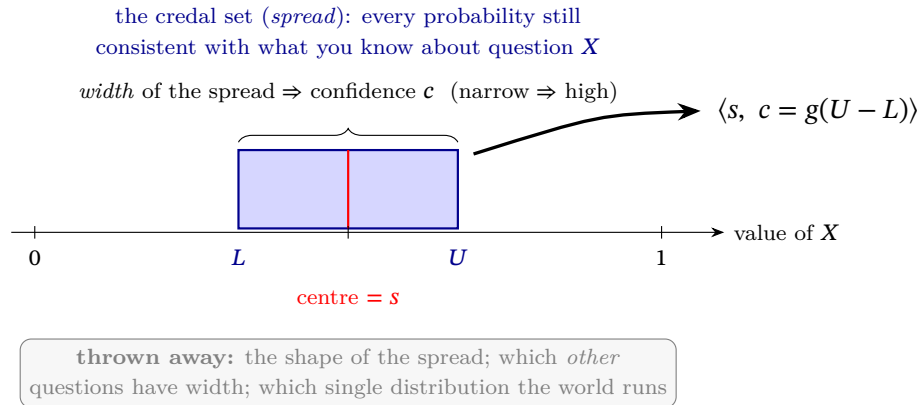


Figure 2: Compression. The two numbers are a lossy photo of a whole set: strength is the centre of the interval the set allows for your question, confidence falls with its width. Everything in the grey box is invisible to $\langle s, c \rangle$.

4 The fork in the road: pinned, or a range

Everything above leads to one clean dichotomy, and it is the technical heart of the paper. Fix a spread and a question X . Exactly one of two things happens.

- **Pinned.** Every admissible distribution in the spread assigns X the *same* expected value. Then $L = U$: the interval is a single point, the width is zero, and confidence sits at its ceiling. The two numbers tell the whole story — there is nothing else to know about X .
- **A genuine range.** Two admissible distributions *disagree* about X . Then $L < U$: the interval has real width, strength is its midpoint, and confidence is honestly below the ceiling. The two numbers report the range, not a single value.

These are the only two cases, and they are mutually exclusive: a question cannot be both pinned and have width (Figure ??). This is stated formally, and machine-checked, in Appendix ??³ (the *pinned/range dichotomy* and its mutual exclusion).

A “pinned” example. If the spread is a single distribution, *everything* is pinned: every question has $L = U$, every confidence is at the ceiling. There is no freedom anywhere — the belief is as sharp as belief can be.

A “range” example: two stuck magnets. Imagine two coupled sites that must agree: either both “up” or both “down,” and you do not know which world you are in. The spread is the two distributions {all-up}, {all-down}. Ask: *is the first site up?* In the all-up world the answer is 1; in the all-down world it is 0. So the interval is $[0, 1]$, width 1, confidence 0: maximally uncertain, even though each individual world is itself perfectly definite. The uncertainty is not noise inside a world — it is not knowing which world.⁴

5 Building a big belief from small pieces

So far the spread was small enough to picture. For a large or infinite world — say, a belief about a whole grid of interacting variables — you cannot write the spread down directly. Instead you specify it *locally*: you say what is admissible on each small “window” of the world, and you glue the windows together consistently. The global spread is then everything that fits all the local pieces at once.

You do not need the machinery to feel the two extremes:

³Lean: `credalSetDetermines` and `credalSetHasStrictWidth` (with the collapse, width, and mutual-exclusion theorems) in [Mettapedia/ProbabilityTheory/ImpreciseProbability/ProjectiveCredal.lean](#); see Appendix ??, item ??.

⁴The two-magnets witness: `twoSpinMagnetEnvelope_nontrivial` in [Mettapedia/ProbabilityTheory/ImpreciseProbability/ProjectiveCredal.lean](#); see Appendix ??, item ??.

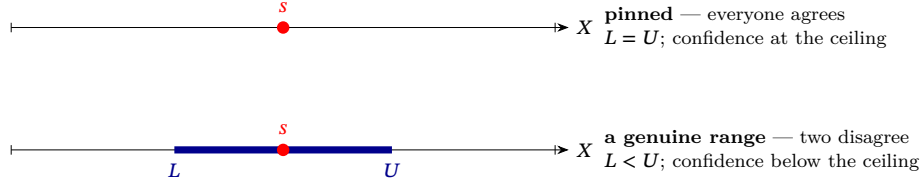


Figure 3: The fork. For any question of any spread: either every admissible distribution agrees (a single pinned value, top) or two disagree (a genuine interval, bottom). This same picture returns for the infinite cases in Section ??: weak coupling lands on top, strong coupling on the bottom. (s is drawn at the midpoint convention.)

- **Tightest possible.** If each local piece allows exactly one distribution, the glued global spread is a single distribution: everything is pinned, no freedom (Section ??, “pinned”).
- **Loosest possible.** If each local piece allows *everything*, the global spread contains every distribution: almost every question has full width, maximum freedom.

Real models live between these. The interesting content is: *where* does freedom enter as you build up, and which kinds can the two-number summary see? That is the next section. (The formal construction — local windows, restriction maps, the glued limit — is Appendix ??, items ?? and following; the body needs only “specify locally, glue globally.”)

6 Three places freedom can hide

Pulling it together, the wiggle room behind $\langle s, c \rangle$ comes from three sources (Figure ??).

- (A) **The modeller’s freedom.** *How wide you chose to make the local pieces.* If you declare a quantity known only to within $[0.4, 0.6]$, you put that width there on purpose. The two numbers report it faithfully: this freedom is **visible**.
- (B) **The world’s freedom.** *Which single true distribution the world actually runs, inside your spread.* Your spread might allow many; the world picks one. The pair $\langle s, c \rangle$ cannot see which — and that is correct, because the belief is a statement about the whole spread, not a guess at the world’s secret pick. This freedom is **invisible**, by design. (It is the kind of uncertainty that more local data does not remove: it is about which possibility is real, not about a fuzzy measurement.) A truth value that *changed* with the world’s hidden pick would be claiming knowledge it cannot have from the evidence — so its blindness here is soundness, not a defect.

(C) **The structure’s own freedom.** *When gluing infinitely many local pieces admits several global solutions to the very same local rules.* No modeller put it there; no single “true world inside the spread” explains it away. It is produced by the structure itself, and the two numbers *can* see it (it shows up as genuine width). This is the surprise of the infinite case (Section ??).

Observation 1 (the thesis in one line). $\langle s, c \rangle$ is a property of the whole spread, not of any single distribution in it. So it sees how wide you made the model (A) and whether the structure forces a range (C), but it is — correctly — blind to which member the world actually picked (B).

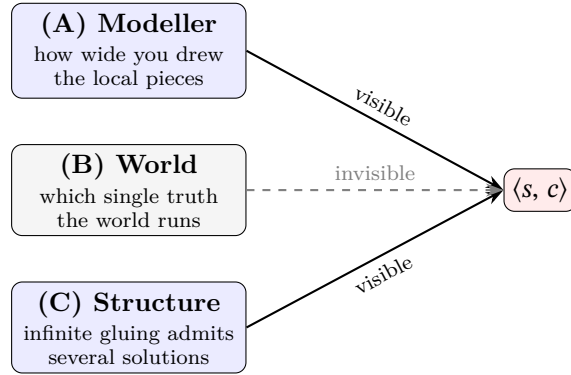


Figure 4: Three sources of freedom and what the two numbers can see. (A) and (C) reach $\langle s, c \rangle$; (B), the world’s single hidden pick, does not — and that blindness is the correct behaviour, because the truth value describes the set, not the pick. (A) is freedom you *declared*; (C) is freedom the structure produces *emergently* — different origins, both visible.

7 Three infinite stories

The same fork — pinned or a range — recurs when the world is infinite, and this is where it gets genuinely interesting. We tell three stories in plain language; the theorems behind them are in Appendix ??.

7.1 When do snapshots add up to a movie? (order-free streams)

You watch a stream where only the *counts* matter, not the order — patient responses, defects coming off a line, fraud flags. “Order does not matter” is a real assumption with a famous name (*exchangeability*), and a beautiful classical fact (de Finetti’s theorem [?]) says such a stream behaves exactly like repeatedly flipping a coin of *unknown* bias, averaged over your uncertainty about the bias.

From any finite prefix you can always write down a sensible next-step forecast. The textbook one is Laplace’s rule: 3 successes in 5 predicts $\frac{3+1}{5+2} = 0.5714$ for the next; 30 in 50 predicts 0.5962. That forecast is *always available*. The subtle part is the stronger claim that your finite forecasts knit together into *one* consistent infinite law that answers *every* question coherently. That stronger claim is not automatic: it holds exactly at a precise compatibility boundary (Appendix ??, item ??).⁵

Takeaway. Having a forecast for tomorrow is free; having a coherent theory of the whole future is not — and we pin down exactly where the line is. A caution that survives the formalization: it is tempting to require the unknown bias to be pinned to an exact value (always 0 or always 1, never in between), but that would forbid the honest “I’m not sure of the bias” case — which is the whole reason to carry a set of coins instead of one. The real boundary is more permissive than that.

7.2 Weak coupling: one precise answer

Now picture not one variable but a vast grid of them — sensors, pixels, people — where each one only *nudges* its neighbours a little. “A little” can be made into a budget you check by adding up the nudges.

The result. When every local influence is small enough (the budget holds), the infinite system has *exactly one* global behaviour consistent with the local rules. Every finite question then gets one precise answer: strength is a single point, confidence sits at the ceiling. Distant conditions wash out — what you fix far away cannot reach the centre. This lands on the *top* of the fork in Figure ??.

Intuitively: if nothing pushes hard, faraway stuff does not matter, so the two numbers tell the whole story. (This is the de-jargoned form of the classical Dobrushin uniqueness condition [?]; Appendix ??, item ??.⁶ We prove this direction; we do *not* claim the converse, that failing the budget forces a range.)

7.3 Strong coupling: a genuine interval the distant edges control

Same grid, opposite regime: now neighbours pull *hard* to agree — a magnet near freezing, a spreading contagion, an image whose adjacent pixels strongly match.

⁵Lean: the *sharp compatibility* crown (and its refuted “every component degenerate” tightening) in [Mettapedia/Logic/DeFinettiProjectiveCredalBridge.lean](#); full name and statement in Appendix ??, item ??.

⁶Lean: `dLrQueryDetermined_of_uniform` (with the precise-envelope and global-uniqueness theorems) in [Mettapedia/Logic/MarkovLogicInfiniteCredalBridge.lean](#).

The result. Now the infinite system has *more than one* global behaviour consistent with the same local rules, and — the striking part — *what you fix at the far-away boundary changes the answer in the centre*. A question at the middle gets a *genuine interval*, not a point: confidence is honestly below the ceiling, and that width is not ignorance you could erase with more local data. It is the system’s own structure. This lands on the *bottom* of the fork.

This is the same setup as Section ?? at a different coupling strength: *weak coupling gives a point, strong coupling gives an interval*. That single contrast is the whole infinite story. Why does the far-away boundary reach the centre at all? Because flipping it would force a long ring of neighbours to disagree, and long rings of disagreement are exponentially unlikely when the coupling is strong — so the centre keeps a memory of the edge that no amount of interior data can erase. The strong-coupling side is the hardest theorem in the project — it is a low-temperature two-dimensional magnet whose centre genuinely depends on the boundary — and it is stated, with the one machine-checked guarantee it provides, in Appendix ??, item ??.⁷ (For the connoisseur: the underlying tool is a Peierls contour argument; we do not need it here, and nothing in this section depends on the proof technique.)

8 The knob zoo: nineteen places two numbers can fool you

This is the practical payoff, and the part to keep open while you work. Each entry is a *degree of freedom*: a modelling choice under which two beliefs can share the same $\langle s, c \rangle$ yet mean different things, or under which a single “obvious” default silently commits you to something. Each comes as: the idea in a sentence, the smallest number that shows it, and the trap it prevents. Every number here is reproduced by the runnable pack (Appendix ??); every entry names its formal anchor in Appendix ??.⁸

1. **Evidence scale κ .**⁹ How much evidence counts as “a lot” is a domain choice. The same $N = 5$ gives confidence 0.833/0.333/0.091 at $\kappa = 1/10/50$.
Trap: fixing $\kappa = 1$ everywhere lets a lone weak hit claim 0.83 confidence

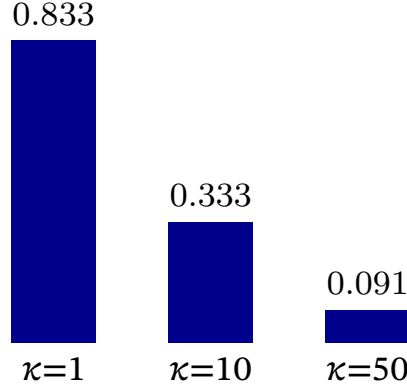
⁷Lean: the symmetric-grid strict-interval crown, proved unconditionally, in [Mettapedia/Logic/MarkovLogicInfiniteSymmetricGridExample.lean](#); full name and statement in Appendix ??, item ??.

⁸The new canaries for DoF 8, 11, and 18 are in [Mettapedia/Logic/PLNConfidenceDoFCanary.lean](#); [papers/pln-confidence-examples/lean/DOF-LEAN-MAP.md](#) maps every degree of freedom to its theorem.

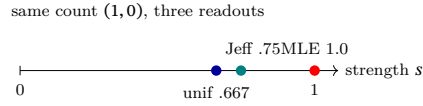
⁹Lean: `walley_width_complement_forces_plnOdds` forces the odds shape while leaving κ free ([Mettapedia/Logic/PLNConfidenceWeight.lean](#)); strength is `posteriorMeanStrength` ([Mettapedia/Logic/PLNTruthTower.lean](#)).

it never earned.

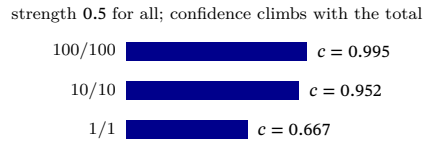
confidence falls as κ rises
(same $N = 5$)



2. **Prior context for strength.**¹⁰ The same counts read differently under maximum-likelihood, a uniform prior, or Jeffreys'. From a single positive (1,0): MLE 1.0, uniform-prior mean 0.667, Jeffreys 0.75 — and all converge to ≈ 0.999 once 1000 observations pile up. *Trap*: pretending strength is context-free silently bakes in a prior.



3. **Total versus ratio.**¹¹ Strength reads the ratio, confidence the total — the reason there are two numbers at all. 1/1, 10/10, 100/100 all have strength 0.5 with confidence 0.667/0.952/0.995; 1/1 versus 2/0 share the total (confidence 0.667) but split on strength (0.5 vs 1.0). *Trap*: any one-number truth value equates 1/1 with 100/100.



4. **Confidence chart.**¹² Recovering the evidence count needs only *some* invertible scale; it does not force PLN's particular one. The same recovered

¹⁰Lean: `posteriorMeanStrength_improper_eq_mle` in [Mettapedia/Logic/PLNTruthTower.lean](#).

¹¹Lean: `strength_mem_unit_of_total_ne_zero` in [Mettapedia/Logic/PLNTruthTower.lean](#).

¹²Lean: `reserveHalfCoordinate_not_valley_width_complement` (a chart that reconstructs evidence yet is not the odds chart) in [Mettapedia/Logic/PLNConfidenceWeight.lean](#).

evidence displayed three ways gives **0.833** (PLN odds), **0.455** (a reserve-half chart), **0.874** (an arctangent chart). *Trap*: combining evidence in a display chart instead of in evidence-count space.

Method	Value
same evidence, three display charts	0.874
arctan	0.874
reserve-half	0.455
PLN odds	0.833

5. **Width-complement convention.**¹³ Setting confidence = 1 – width is natural but still a convention, and it needs a scale. A binary interval gives 0.4; a bounded utility, correctly range-normalized, also 0.4; the same utility left un-normalized gives the deliberate nonsense –59.0. *Trap:* applying width-complement to dollar-valued gambles without normalizing.

width 0.6 $\Rightarrow c = 1 - 0.6 = 0.4$

but an un-normalized utility gives $c = -59$ (nonsense)

6. **Interval point selector.**¹⁴ A real interval $[L, U]$ does not pick its own display point. Lower / midpoint / upper on one interval: 0.2/0.5/0.8. *Trap:* a safety decision shown the midpoint when it needed the lower bound.

A horizontal line segment representing the interval [0, 1]. The left endpoint is labeled '0' and the right endpoint is labeled '1'. Above the line, three points are marked with red dots. The first dot is at 0.2 and is labeled 'lower .2' in red. The second dot is at 0.5 and is labeled 'mid .5' in red. The third dot is at 0.8 and is labeled 'upper .8' in red. A thick blue line segment connects the first and third dots. Below the line, the text 'one interval, three honest picks' is written.

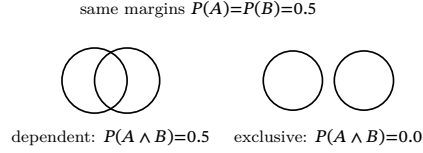
7. **Credal shape beyond one interval.**¹⁵ Two spreads can agree on one question and disagree on another — the very reason $\langle s, c \rangle$ is a compression. Two joint models with identical marginals $P(A) = P(B) = 0.5$ can have conjunction 0.5 (perfectly dependent) or 0.0 (mutually exclusive); the whole range $[0, 0.5]$ is invisible to the marginal summary. *Trap:* certifying two incompatible joint models “the same” because their margins match. A second, operational source of the same width is unresolved observational

¹³Lean: `WidthComplementCompatible` ([Mettapedia/Logic/PLNTruthTower.lean](#)) and `walley_width_complement_forces_plnOdds` ([Mettapedia/Logic/PLNConfidenceWeight.lean](#)).

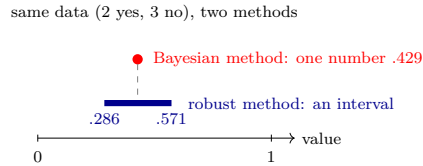
¹⁴Lean: `generic_itv_does_not_force_point_projection` in [Mettapedia/Logic/PLNTruthTheoryIndex.lean](#).

¹⁵Lean: `CredalProjectionTower` in `Mettapedia/Logic/PLNTruthTower.lean`; `observationCredalSet_indicatorGamble_has_strict_width_of_related_ne` and `observationCredalSet_indicatorGamble_collapses_of_class_subsingleton` in `Mettapedia/Logic/PLNDistinctionCredalBridge.lean`; and the OSLF specialization `indistObs_indicatorGamble_has_strict_width / indistObs_indicatorGamble_collapses_of_class_subsingleton` in `Mettapedia/Logic/PLNDistinctionCredalOSLFBridge.lean`.

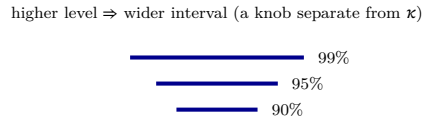
distinction: if the observer cannot yet tell two states apart, the relevant truth query can remain interval-valued until that distinction is resolved.



8. **Which uncertainty method (the “backend”).**¹⁶ Feed the *same* data to two different uncertainty methods and you get answers of different *shape*. From 2 “yes” and 3 “no”: a single-distribution Bayesian method returns one best number (the Beta posterior mean $\frac{3}{7} \approx 0.429$), while a cautious *imprecise* method returns a whole interval of defensible values, $[\frac{2}{7}, \frac{4}{7}] = [0.286, 0.571]$. The Bayesian point lands inside that interval — it simply declines to show the range. *Trap*: collapsing the cautious interval down to the single point “for convenience,” silently discarding the safety margin.



9. **Credible level / tolerance.**¹⁷ A Bayesian interval also carries a chosen level (90/95/99%) — a different knob from κ . κ governs how evidence accumulates; the level governs what you *ask* of a finished posterior. *Trap*: confusing “more evidence” with “a tighter requested interval.”



10. **Categorical query.**¹⁸ In a multi-class model the total evidence is shared but each category has its own confidence. In a 1000-sample 3-class model: the common class has strength 0.90, confidence 0.999; a rare class has strength 0.002, confidence 0.167 *despite* the huge total. *Trap*: “the classi-

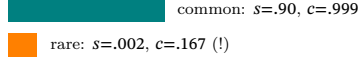
¹⁶Lean: `beta_mean_strictly_inside_walley_idm_interval` in [Mettapedia/Logic/PLNConfidenceDoFCanary.lean](#); see Appendix ??.

¹⁷Lean: `walleyCategorical_credibility_eq` in [Mettapedia/Logic/PLNTruthTower.lean](#).

¹⁸Lean: `categoricalSurface_idmEnvelope_forced_by_aggregate` in [Mettapedia/Logic/PLNTruthTower.lean](#).

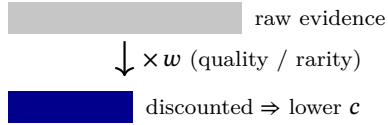
fier is 99.9% confident” hiding that a class rests on two observations.

one model, very different per-class confidence

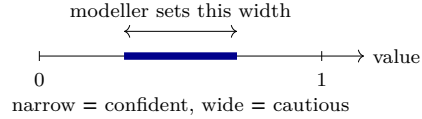


11. **Evidence discounting.**¹⁹ Evidence can be reweighted for quality, rarity, or route before it counts. Power-discounting scales the log-odds linearly; discounted totals of 4.57, 12, 104 give confidences 0.821/0.923/0.991. *Trap*: treating correlated hits as independent and overstating confidence.

down-weight before counting

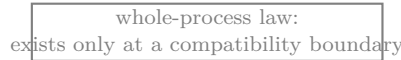


12. **Local credal specification.**²⁰ In a built-up model the modeller chooses how wide each local piece is — the intended-imprecision knob (source (A) of Section ??). *Trap*: hiding local ambiguity behind a single precise prior, so later confidence looks unjustifiably high.



13. **Compatible completion.**²¹ Finite-prefix forecasts always exist; a full infinite law does not always exist (Section ??). The Laplace readout $3/5 \rightarrow 0.5714$, $30/50 \rightarrow 0.5962$ is always available; the consistent whole-process law is a separate boundary condition. *Trap*: calling finite forecasts a process law without the compatibility witness.

$3/5 \longrightarrow$ next ≈ 0.571 (always free)



14. **Completion extreme point.**²² Inside a fixed spread the world runs

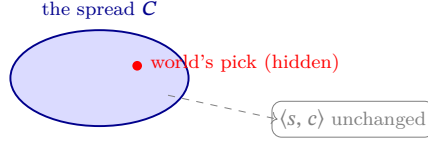
¹⁹Lean: `power_discount_scales_logodds` in [Mettapedia/Logic/PLNConfidenceDoFCanary.lean](#).

²⁰Lean: `finiteLowerPrevision` and `coherentDesirableSet` in [Mettapedia/Logic/PLNTruthTower.lean](#).

²¹Lean: the sharp-compatibility crown in [Mettapedia/Logic/DeFinettiProjectiveCredalBridge.lean](#); see Appendix ??, item ??.

²²Lean: `boundedMeasurableCredalSetHasStrictWidth` in [Mettapedia/ProbabilityTheory/ImpreciseProbability/ProjectiveCredal.lean](#).

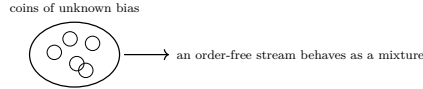
one true distribution, and that single pick is invisible to $\langle s, c \rangle$ — correctly (source (B)). The two stuck magnets witness it: two admissible worlds disagree, so no single determined readout is justified. *Trap*: an inference rule that pretends to know which world is real.



15. **Phase transition / infinite boundary.**²³ Some infinite specifications admit several global solutions to the same local rules (source (C)). Weak coupling collapses the interval (Section ??); strong coupling keeps it open, controlled by distant boundaries (Section ??). *Trap*: assuming a unique global model exists just because the local rules are fixed.



16. **Exchangeability / mixture readout.**²⁴ An order-free stream has a canonical mixture forecast, but the raw all-questions law depends on the compatibility boundary of Section ?. *Trap*: demanding every mixture component be degenerate, which wrongly rejects legitimate “unsure-of-the-bias” families.



17. **Rule-level confidence transport (revision).**²⁵ Combining two independent bodies of evidence is “add the counts, then re-read,” not “average the confidences.” $(4, 2) \oplus (3, 1) = (7, 3)$ gives $\langle 0.7, 0.909 \rangle$ — higher than either source ($\langle 0.667, 0.857 \rangle$ and $\langle 0.75, 0.800 \rangle$). *Trap*: averaging displayed

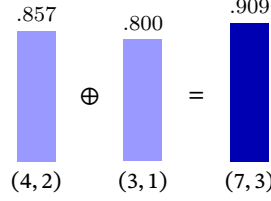
²³Lean: the weak-coupling collapse ([Mettapedia/Logic/MarkovLogicInfiniteCredalBridge.lean](#)) and the strong-coupling crown ([Mettapedia/Logic/MarkovLogicInfiniteSymmetricGridExample.lean](#)); see Appendix ??, items ?? and ??.

²⁴Lean: the imprecise de Finetti mixing-family in [Mettapedia/Logic/DeFinettiProjectiveCredalBridge.lean](#); see Appendix ??, item ??.

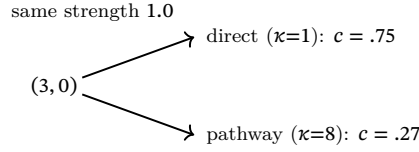
²⁵Lean: `transportedConfidenceRevision` and `plnConfidenceRevision` in [Mettapedia/Logic/PLNConfidenceWeightRevision.lean](#).

confidences, which absurdly *lowers* confidence after seeing *more* evidence.

add counts $\Rightarrow$ higher c (not the average!)

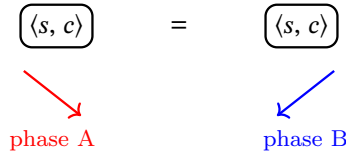


18. **Provenance / route.**²⁶ Same counts, different route, route-specific confidence. From $(3, 0)$: strength 1.0 on every route, but confidence 0.75 on a direct route ($\kappa = 1$) versus 0.273 on a pathway-mediated route ($\kappa = 8$). *Trap*: merging routes before applying route-specific κ , destroying both the audit trail and the discount.



19. **Extra structure beyond $\langle s, c \rangle$.**²⁷ Some beliefs carry more than strength and confidence — a “phase” or dependence witness — and two beliefs with identical $\langle s, c \rangle$ can differ once a composition rule that *sees* the extra coordinate is in play. *Trap*: both carrying phase needlessly in a plain classifier, *and* assuming $\langle s, c \rangle$ is complete when interference-like composition matters.

same pair, different hidden coordinate



The thread through all nineteen: a degree of freedom is a place where a *convention* masquerades as a *fact*. Naming them is what lets you choose deliberately — and lets two honest modellers reconcile beliefs that print the same two numbers.

²⁶Lean: `route_kappa_changes_confidence_not_strength` in [Mettapedia/Logic/PLNConfidenceDoFCanary.lean](#).

²⁷Lean: `phase_not_visible_to_standard_pln_view` in [Mettapedia/Logic/PLNAmplitudePhase.lean](#).

9 Conclusion

PLN’s truth value is two numbers standing in for a whole set of admissible probabilities. Strength is the centre of the set on the question you ask; confidence falls with the set’s width. The two numbers tell the whole story exactly when the set pins your question to a point, and report a genuine interval otherwise — one fork, recurring from a single coin up to infinite interacting systems. Freedom hides in three places: the width you chose (visible), the world’s single hidden pick (invisible, and rightly so), and the structure’s own non-uniqueness in the infinite limit (visible, as real width). And nineteen everyday modelling choices live in the gap between “the two numbers” and “the set behind them.” Every claim here has a machine-checked counterpart (Appendix ??) and a runnable demonstration (Appendix ??).

10 Related work

Imprecise probability. The credal-set / lower-prevision picture — a belief as a *set* of distributions, summarized by lower and upper expectations — is Walley’s [?]. Our “spread,” “pinned versus range,” and the read-off of strength and confidence are the PLN-facing form of his coherent lower previsions and their envelopes.

Exchangeability. de Finetti’s theorem [?] is the classical bridge: an exchangeable (order-free) infinite sequence is a mixture of i.i.d. trials. Section ?? is the imprecise version — a *set* of such mixtures, with the compatibility boundary that decides when finite forecasts extend to one whole-process law.

Infinite systems. The Dobrushin–Lanford–Ruelle framework [?, ?] characterizes infinite Gibbs measures by their finite-volume conditionals; Dobrushin’s uniqueness condition is the “weak coupling, one answer” result of Section ??, and phase-transition non-uniqueness is the “strong coupling, an interval” result of Section ?. Markov logic networks [?, ?] are the logic-facing instance: weighted clauses specifying a distribution over possible worlds.

Sibling papers. This note is a companion to §PLN and the PLN Review ([papers/xiPLN.tex](#), [papers/pln-review.tex](#)), which treat the evidence algebra and PLN’s inference rules; the WM-PLN book [?] contains the informal predecessor of the degrees-of-freedom discussion this paper makes precise.

A Run it yourself

Every number in the body is reproduced by a small runnable companion pack at [papers/pln-confidence-examples/](#). It runs on two independent MeTTa

engines (CeTTa in its HE mode, and PeTTa), so the arithmetic is not taking any one implementation’s word for it.

```
cd papers/pln-confidence-examples
sh run_all.sh          # build CeTTa, run every dofNN, then PeTTa
```

The pack has one small example per degree of freedom (`metta/dof01..18`), each a self-contained file ending in queries whose printed outputs are the claims; a one-line-per-knob index (`README.md`); and the expected numbers for a human spot-check. The honest contract is stated there and bears repeating: **the Lean theorems of Appendix ?? are the correctness oracle; the `.metta` runs are intuition and reproducibility, never proof.** Matching output is an illustration check, not a verification.

A few entries (the infinite-boundary ones — exchangeable streams, weak/strong coupling, completion extreme points, hidden phase) have no finite executable that could settle the claim; for those the pack ships the closest finite “shadow” (e.g. the Laplace forecast of Section ??) and points at the theorem that carries the real content. The index marks each row as *runnable*, *structural* (a finite shadow), or *theorem-only*, so nothing infinite masquerades as something you can compute.

B Formal companion: the statements, de-jargoned

The body avoided machinery on purpose. Here are the load-bearing statements, each as: the plain-English claim the body relied on, the machine-checked Lean name, and one line on what it forces. All are verified in the Mettapedia Lean library; the public entry point bundling them is `confidenceCharacterizationEndpointProfile`, and the names below are live declarations in that development. The core objects: a *gamble* is a real-valued function on the outcome space (a random variable / payoff); a *precise prevision* is an expectation operator $E[\cdot]$ on gambles; a *credal set* (our “spread”) is a set of these; the *lower* and *upper envelopes* are the pointwise inf and sup of $E[X]$ over the set — the endpoints L and U . The listings are lightly ASCII-formatted (Ω for Ω , $\rightarrow$ for $\rightarrow$, and `forall/exists/in/and/not` for the binders and connectives). The source files are [Mettapedia/ProbabilityTheory/ImpreciseProbability/ProjectiveCredal.lean](#) (the credal stack), [Mettapedia/Logic/DeFinettiProjectiveCredalBridge.lean](#), [Mettapedia/Logic/MarkovLogicInfiniteCredalBridge.lean](#), and [Mettapedia/Logic/MarkovLogicInfiniteSymmetricGridExample.lean](#).

1. **The read-off.** Strength is the midpoint and confidence a decreasing function g of the width of $[L, U]$. The midpoint and g are interface choices; none of the results below depends on them. *Forces*: the meaning of $\langle s, c \rangle$ as centre-and-width of an envelope interval.

```

abbrev CredalPrevisionSet (Omega : Type*) := Set (
  PrecisePrevision Omega)

def lowerEnvelope (C : CredalPrevisionSet Omega) (X : Gamble
  Omega) : Real :=
  sInf ((fun P : PrecisePrevision Omega => P X) '' C)
def upperEnvelope (C : CredalPrevisionSet Omega) (X : Gamble
  Omega) : Real :=
  sSup ((fun P : PrecisePrevision Omega => P X) '' C)

```

2. **The pinned/range dichotomy (the spine).** If every admissible prevision assigns X the same value the envelopes coincide; if two disagree the envelope strictly opens; and the two cases are mutually exclusive. *Forces:* Section ?? — pinned and range are the only cases and cannot coexist.

```

def credalSetDetermines (C : CredalPrevisionSet Omega) (X :
  Gamble Omega) : Prop :=
  forall P : PrecisePrevision Omega, P in C ->
    forall Q : PrecisePrevision Omega, Q in C -> P X = Q X
def credalSetHasStrictWidth (C : CredalPrevisionSet Omega) (X :
  Gamble Omega) : Prop :=
  exists P : PrecisePrevision Omega, P in C and
    exists Q : PrecisePrevision Omega, Q in C and P X < Q X

theorem not_credalSetDetermines_of_strictWidth
  (hWidth : credalSetHasStrictWidth C X) : not (
    credalSetDetermines C X)

theorem lower_eq_upperEnvelope_of_credalSetDetermines
  (C : CredalPrevisionSet Omega) (X : Gamble Omega) (hC : C.
    Nonempty)
  (hBddBelow : BddBelow ((fun P => P X) '' C))
  (hBddAbove : BddAbove ((fun P => P X) '' C))
  (hP : P in C) (hDet : credalSetDetermines C X) :
  lowerEnvelope C X = upperEnvelope C X

theorem lower_upperEnvelope_nontrivial_of_strictWidth
  (C : CredalPrevisionSet Omega) (X : Gamble Omega)
  (hBddBelow : BddBelow ((fun P => P X) '' C))
  (hBddAbove : BddAbove ((fun P => P X) '' C))
  (hWidth : credalSetHasStrictWidth C X) :
  lowerEnvelope C X < upperEnvelope C X

```

3. **Finite totality.** On a finite outcome space the construction never misbehaves — the spread avoids sure loss and is coherent — and a one-element spread pins every question. *Forces:* the map from a specification to $\langle s, c \rangle$ is well defined.

```

theorem lowerEnvelopePrevision_avoidsSureLoss_of_finite
  [Fintype Omega] [Nonempty Omega]
  (C : CredalPrevisionSet Omega) (hC : C.Nonempty)
  (hBdd : forall X, BddBelow ((fun P => P X) '' C)) :
  (lowerEnvelopePrevision C hC hBdd).avoidsSureLoss

```

```

theorem lowerEnvelope_singleton (P : PrecisePrevision Omega) (X
  : Gamble Omega) :
  lowerEnvelope ({P} : CredalPrevisionSet Omega) X = P X

```

4. **The dichotomy lifts to built-up specs.** For beliefs glued from local windows, the same fork holds globally. *Forces:* Section ??.

```

def determinesGlobalGamble
  (S : ProjectiveLocalCredalSpec Window Global) (X : Gamble
    Global) : Prop :=
  credalSetDetermines S.projectiveLimitCredalSet X
def hasStrictGlobalWidth
  (S : ProjectiveLocalCredalSpec Window Global) (X : Gamble
    Global) : Prop :=
  credalSetHasStrictWidth S.projectiveLimitCredalSet X

theorem not_determinesGlobalGamble_of_strictWidth
  (S : ProjectiveLocalCredalSpec Window Global)
  (hWidth : S.hasStrictGlobalWidth X) : not (S.
    determinesGlobalGamble X)

```

5. **Order-free streams.** The canonical mixture forecast always exists; the stronger “one consistent law for every question” holds *exactly* when a respecting lower prevision exists. It may *not* be tightened to “every mixture component degenerate” — that strengthening is refuted. *Forces:* the take-away and caution of Section ??.

```

theorem impreciseDeFinetti_analyticMixingFamily_
  sharpCompatibilityCrown
  (C : Set BernoulliMixture) (hC : C.Nonempty) :
  ImpreciseDeFinettiCanonicalExternalMixingFamily C hC and
  (ImpreciseDeFinettiAnalyticMixingFamilyProcessLawCrown C
    hC <->
    exists L : LowerPrevision (Nat -> Bool),
      (bernoulliMixturePrefixProcessLowerSpec C
        (fun M _ n => bernoulliMixturePrefixLaw_analytic M n
          ) hC).respectsLocalLower L)

theorem not_forall_impreciseDeFinetti_
  analyticMixingFamilyProcessLawCrown_imp_
  pointwiseZeroInterior_closed :
  not (forall (C : Set BernoulliMixture) (hC : C.Nonempty),
    ImpreciseDeFinettiAnalyticMixingFamilyProcessLawCrown C
      hC ->
      AnalyticMixingFamilyPointwiseZeroInterior C)

```

6. **Weak coupling collapses.** Under a uniform small-influence budget every finite query is pinned, and the global model is unique. We prove only this direction; the converse (budget failure forcing width) is not claimed. *Forces:* Section ??, top of the fork.

```

theorem dlrQueryDetermined_of_uniform
  (M : ClassicalInfiniteGroundMLNSpec Atom ClauseId)

```

```

(hM : M.PaperUniformSmallTotalInfluence) (q :
  ConstraintQuery Atom) :
  dlrQueryDetermined M q

theorem infiniteMLN_queryEnvelope_precise_of_uniform
  (M : ClassicalInfiniteGroundMLNSpec Atom ClauseId)
  [Nonempty (DLRCompletion M)]
  (hM : M.PaperUniformSmallTotalInfluence) (q :
    ConstraintQuery Atom) :
  infiniteMLNLowerQueryEnvelope M q =
    infiniteMLNUpperQueryEnvelope M q

```

7. **Strong coupling keeps an interval.** A low-temperature two-dimensional symmetric grid has an origin query with genuinely positive envelope width (so confidence is strictly below its ceiling), a range the distant boundary controls — and this is proved with *no* hypotheses. The two stuck magnets are the finite witness used in the body. *Forces*: Section ??, bottom of the fork.

```

-- the crown asserts, for the zero-field grid at coupling w:
structure SymmetricGridZeroFieldOriginPLNStrictIntervalCrown
  (w : Real) : Prop where
  queryEnvelopeWidth_pos :
    0 < infiniteMLNQueryEnvelopeWidth
      (symmetricGridZeroFieldClassicalSpec w)
      gridOriginSpinUpQuery
  queryEnvelopeWidthComplement_lt_one :
    infiniteMLNQueryEnvelopeWidthComplement
      (symmetricGridZeroFieldClassicalSpec w)
      gridOriginSpinUpQuery < 1
  queryOutcomeCredalSet_width_pos : ... -- the binary outcome
    has positive width

-- proved with no hypotheses at coupling w = 24:
theorem symmetricGridZeroField_originPLNStrictIntervalCrown_of_
  axisAnchoredContourCode_twentyFour :
  SymmetricGridZeroFieldOriginPLNStrictIntervalCrown (24 :
    Real)

-- the finite witness used in the body (two stuck magnets):
theorem twoSpinMagnetEnvelope_nontrivial :
  lowerEnvelope twoSpinMagnetCredalSet twoSpinFirstUpGamble <
    upperEnvelope twoSpinMagnetCredalSet twoSpinFirstUpGamble

```

8. **Three freedoms, formally.** The sharpest negative claim of the paper: $\langle s, c \rangle$ is a property of the credal *set*, so the world's single choice of member (source (B)) cannot change it, while the spec width (A) and structural non-uniqueness (C) can. This is the content of items ??–?? read together, and is why source (B) is invisible in Figure ??.

The full type definitions and proof scripts (the credal stack, the projective construction, the DLR specialization, and the contour argument behind item ??) live in the Lean sources under `lean/mettapedia/Mettapedia/`; this

appendix states what they establish, not how. The companion package’s `lean/DOF-LEAN-MAP.md` maps each degree of freedom in Section ?? to its specific theorem.

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